Kyoko MATSUI

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Research Profile

I study procurement systems and contractual governance for large, technically complex public projects. My research uses game theory, contract theory and comparative institutional analysis to examine investment in specifications, vendor lock-in, contract changes and dispute resolution. I bring over eight years of experience at Mitsubishi Research Institute and Dassault Systèmes, spanning defence systems procurement support, requirements analysis in aerospace and manufacturing, and public project appraisal.

Research Interests

Public procurement and contract design; early contractor involvement; incomplete contracts and investment incentives; contract changes and dispute resolution; applications to defence procurement and international industrial cooperation.

Education

- **Oct 2024–present | Kyoto University, Graduate School of Engineering**
  Doctoral programme, Department of Urban Management. PhD expected September 2027. Supervisor: Professor Masamitsu Onishi.
- **Apr 2014–Mar 2016 | The University of Tokyo, Graduate School of Engineering**
  Master of Engineering, Department of Civil Engineering. Research on landslide warning policy under precipitation forecast uncertainty and disaster risk.
- **Apr 2010–Mar 2014 | The University of Tokyo**
  Bachelor of Engineering, Department of Civil Engineering, following the university's junior division programme.

Research Appointment and Funding

- **Apr 2026–present | JSPS Research Fellow DC2**
  Fellowship period: April 2026–March 2028. Principal investigator, JSPS Grant-in-Aid for JSPS Fellows, Theoretical Study on Procurement System for Large-Scale Public Works. Project number 26KJ1413

Peer Reviewed Publications

1. **Matsui, K.**, and Onishi, M. (2025). A theoretical study on the competitiveness of the price negotiation process in ECI. *Japanese Journal of JSCE*, 81(20), 25-20016. In Japanese; published online 30 June 2026. doi:10.2208/jscejj.25-20016
2. **Matsui, K.**, and Shimamura, M. (2016). A study of landslide warning policy based on precipitation forecast and risk assessment. *Journal of Japan Society for Natural Disaster Science*, 35 (Special Issue), 25–38. In Japanese. Full text
3. **Matsui, K.**, and Shimamura, M. (2015). Optimal operation of tornado warnings based on cost-loss model. *Journal of Japan Society of Civil Engineers, Ser. D3 (Infrastructure Planning and Management)*, 71(5), I_111–I_119. In Japanese. doi:10.2208/jscejipm.71.I_111

Research in Progress

- **Contractual Governance of Public Works: Evidence Production and Monitoring in the UK and Japan**
  With Masamitsu Onishi. Manuscript in preparation; first and corresponding author. Compares incentives for evidence production and monitoring in UK and Japanese approaches to contract changes and dispute resolution. Intended journal: Group Decision and Negotiation.
- **Incentives for Joint Design Investment in Early Contractor Involvement**
  First and corresponding author. Theoretical analysis of buyer and contractor investment in specifications and the role of remuneration. Conference paper submitted to the 44th JSCE Construction Management Symposium, for presentation in December 2026.

Selected Presentations

- **1 July 2026** | Contractual Governance of Public Works: Evidence Production and Monitoring in the UK and Japan. 26th International Conference on Group Decision and Negotiation, Katowice, Poland. Oral presentation. In English.
- **9 December 2025** | Extending Public Procurement Studies through the Perspective of Institutional Linkage: a Conceptual Review and Outlook. With Masamitsu Onishi. 43rd JSCE Construction Management Symposium. In Japanese.
- **8 June 2025** | A Theoretical Study on the Competitiveness of the Price Negotiation Process in ECI. With Masamitsu Onishi. 71st JSCE Infrastructure Planning and Management Conference, Spring Meeting. In Japanese.

Awards

- **August 2026** | Outstanding Presentation Award, JSCE Construction Management Committee, for research on institutional linkages in public procurement, with Masamitsu Onishi.
- **September 2017** | Academic Encouragement Award, Japan Society for Natural Disaster Science, for research on landslide warning policy, with Makoto Shimamura.
- **September 2016** | Outstanding Presentation Award, Refereed Session, 35th Annual Conference of the Japan Society for Natural Disaster Science, for the same research.

Professional Experience

Business Consultant | Dassault Systèmes K.K.

June 2022–September 2024 | Tokyo, Japan

Pre-sales consulting on business processes, requirements and proposed solutions for manufacturing, aerospace and construction clients.

- Served as the main workshop facilitator for requirements analysis in an aerospace business platform project. Analysed client documents, prepared questions, elicited business processes and organised quality planning and approval requirements using CATIA Magic.
- Took primary responsibility for the consulting component of a construction-sector business transformation engagement, including interviews, analysis of current processes, future process documentation, ROI calculations and presentations to the client.
- Documented future business processes for a consumer-sector service involving 3D scanning and 3D printing, using interviews and CATIA Magic models.
- Prepared consulting proposals and project schedules, analysed interview text and consolidated technical sales input for manufacturing clients.

Researcher | Mitsubishi Research Institute, Inc.

April 2016–May 2022 | Tokyo, Japan

Defence-related systems procurement support and policy research for Japanese central government clients.

- **Systems procurement and technical review:** Supported a communications and command-and-control systems renewal programme in a team of approximately four during 2016–2020. Reviewed specifications and design documents, checked consistency with international standards, and supported tests and reviews. Review comments were incorporated into design and test plans.
- **International procurement and coordination:** Supported overseas equipment procurement through requirements communication, standards review, delivery follow-up, acceptance activities and domestic supplier selection. Assisted government-to-government coordination on joint activities through meeting materials, scheduling and advice.
- **Industrial spillovers from aircraft development:** Conducted interviews with manufacturers and related industries to identify technologies and potential spillovers, and contributed to indicative estimates of economic effects for policy consideration.
- **Supply chain research:** Synthesised existing manufacturer interviews on supply chain sustainability into reference materials for a central government client.
- **Public infrastructure project appraisal:** Supported expert committee work for revisions to a cost-benefit appraisal manual in 2021, preparing materials and coordinating meetings while maintaining consistency with existing guidance and relevant academic knowledge.
- **Facility location research:** Interviewed overseas experts on the local effects of comparable facilities and prepared materials adopted for a developer's internal presentations.

Methods and Skills

- **Theory and institutional analysis:** Game theory, principal–agent models, contract theory and comparative institutional analysis applied to procurement systems.
- **Quantitative research:** Python for data processing and numerical simulation; prior research use of R and SQL. Cost-loss models, value at risk and expected shortfall in risk and decision analysis.
- **Policy research:** Literature and standards reviews, expert and industry interviews, and preparation of policy and technical documents.
- **Requirements and process analysis:** CATIA Magic, stakeholder and requirements analysis, business process modelling, workshop facilitation and ROI calculations.
- **Languages:** Japanese (native); English used in international stakeholder coordination, technical literature reviews, conference presentations and research discussions. TOEFL iBT 101 (February 2023); TOEIC 955 (December 2022).
- **Qualification:** Systems Auditor Examination, Information-technology Promotion Agency, Japan (2022).

Professional Memberships

Japan Society of Civil Engineers; Japan Society for Natural Disaster Science.

Availability

Available from October 2027, subject to completion of the PhD expected in September 2027.